

Next Steps After Discovering Four Clusters

A practical roadmap from unsupervised visual regime discovery to a validated supervised classifier

Current position

The unsupervised pipeline has completed successfully and selected four clusters. The present provisional interpretations are:

Cluster	Provisional interpretation
00	High-volatility consolidation
01	Bullish trend and breakout
02	Volatile sideways and reversal
03	Bearish trend and breakdown

1. Quantitatively validate the clusters

Before finalizing the regime names, calculate market statistics for every cluster using the original OHLC data. Recommended statistics include average and median window return, realized volatility, trend slope, proportion of bullish candles, maximum drawdown, average candle range, and reversal frequency.

The objective is to check whether the numerical behaviour supports the visual labels. For example, Cluster 01 should normally show positive returns and positive trend slopes, while Cluster 03 should show negative returns and negative trend slopes.

2. Check class balance and split consistency

Use `train_cluster_labels.csv`, `validation_cluster_labels.csv`, and `cluster_summary.csv` to compare the number and proportion of samples in each cluster. Check whether one class is substantially larger than the others and whether the validation distribution resembles the training distribution.

If imbalance is substantial, plan to use class-weighted cross-entropy loss, a weighted sampler, or both during supervised training.

3. Finalize the regime labels

Retain the numerical cluster IDs and store the descriptive names separately. A suggested mapping is:

ID	Label
0	high_volatility_consolidation
1	bullish_trend_breakout
2	volatile_sideways_reversal
3	bearish_trend_breakdown

4. Build the supervised classifier

The classifier will learn to map each candlestick image to one of the four cluster IDs. Use `train_cluster_labels.csv` for training and `validation_cluster_labels.csv` for validation. The image path is the input and `cluster_id` is the target.

Component	Recommended choice
Model	Pretrained ResNet-18
Output layer	Four output neurons
Loss	Cross-entropy loss
Optimizer	Adam or AdamW
Checkpoint rule	Save the model with the best validation macro F1-score or validation loss
Augmentation	Use carefully; avoid transformations that change financial meaning

5. Evaluate the classifier

Do not rely on accuracy alone. Report validation accuracy, macro and weighted precision, recall, macro F1-score, weighted F1-score, per-class performance, confusion matrix, and training/validation loss curves. Macro F1 is especially important if the cluster sizes are unequal.

The classifier's role is to reproduce the K-Means pseudo-labels efficiently. High classifier accuracy does not independently prove that the regimes are economically meaningful.

6. Add an untouched chronological test set

For a stronger study, reserve a later chronological period that was not used during PCA fitting, K selection, K-Means fitting, or classifier tuning. Evaluate the final classifier once on this test set to estimate out-of-sample generalization.

7. Compare classifier predictions with K-Means assignments

Measure agreement between the classifier and the original clustering pipeline using accuracy, confusion matrix, adjusted Rand index, and percentage agreement. This demonstrates whether the classifier successfully approximates the embedding-PCA-K-Means pipeline.

8. Test economic usefulness

After classification, analyse the market behaviour associated with each regime. Useful analyses include future returns, future volatility, regime duration, transition probabilities, probability of upward or downward movement, and the performance of simple regime-aware rules.

Recommended sequence

Step	Action
1	Quantitatively validate the four clusters using OHLC data
2	Finalize regime names and check class balance
3	Train the four-class ResNet-18 classifier

4	Evaluate on validation data and an untouched chronological test set
5	Compare classifier outputs with K-Means assignments
6	Analyse regime transitions and future market behaviour
7	Write the final results and discussion chapters

Immediate next task: perform quantitative validation of the four discovered clusters using the original OHLC data. The supervised classifier should be built after the regime labels have been supported or revised by those statistics.